

ADARSH KUMAR SINGH

AI Engineer · Agentic Systems · RAG · LLM Evaluation · 6 Live Production Deployments

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PROFESSIONAL SUMMARY

AI Engineer specializing in production-grade agentic systems, RAG pipelines, and LLM evaluation infrastructure. Built and deployed 6 publicly accessible AI applications — spanning hedge fund intelligence, enterprise compliance, autonomous DevOps orchestration, real-time voice agents, and multimodal Vision AI — all live on Hugging Face and Vercel. Technical Lead at AKGEC Software Incubator with hands-on expertise in LangChain, LangGraph, tool-calling architectures, and LLM observability using RAGAS and Langfuse. Oracle-certified in Generative AI and Data Science. Member: Google Developer Groups · NVIDIA Developer Program.

WORK EXPERIENCE

Technical Lead — AI Engineering, AKGEC Software Incubator

Feb 2023 – Mar 2025

Ajay Kumar Garg Engineering College, Ghaziabad | Full-Time

- Architected a stateful enterprise compliance agent using LangGraph state graphs and ReAct patterns; reduced hallucination rate by 40% via hybrid RAG (dense + BM25) with FAISS — deployed live with I/O guardrails blocking 100% of tested prompt injection attempts.
- Built an LLM-as-a-Judge evaluation pipeline with LangChain, RAGAS, and Langfuse tracking faithfulness, relevance, and context recall in real time; implemented automatic model failover (70B → 8B) cutting post-release defects by 35%.
- Designed an Autonomous AI DevOps Swarm of 5 specialized sub-agents with tool-calling and structured outputs; intercepted CI/CD failures, wrote patches, and opened PRs without human intervention — 94% auto-resolution rate, MTTR from 30+ min to sub-60 seconds.
- Engineered Clara AI, a multimodal clinical receptionist (Groq Vision AI + live TTS + WebSocket sync) for NHS GP triage; deployed on Vercel with Next.js. Built a companion Real-Time Voice Agent achieving sub-500ms TTFB over SSE/WebSocket.
- Shipped SentinelAlpha (SEC EDGAR intelligence), ComplianceAI Enterprise (GDPR/SOX/HIPAA), NeuralHeist (LLM game backend), SmartLedger (Vision AI finance), and PersonaDoc RAG — all publicly deployed on Hugging Face Spaces.
- Delivered infrastructure on AWS (EC2, S3, Lambda) and Oracle Cloud with Docker, Kubernetes, and GitHub Actions CI/CD; maintained 99.5% uptime across all production agents.

KEY PROJECTS

SentinelAlpha — Hedge Fund Intelligence Platform — *Python · FastAPI · Groq · SEC EDGAR · Streamlit · Plotly*

- Analyzes SEC filings, earnings transcripts, and market signals to generate alpha conviction scores for hedge fund analysts. Live on Hugging Face.

Autonomous AI DevOps Swarm — *Python · LangGraph · Multi-Agent · Tool Calling · Docker · GitHub Actions*

- 5-agent system that intercepts CI/CD failures, diagnoses root cause, writes code patches, runs sandboxed tests, and opens PRs — fully autonomously. 94% resolution rate, MTTR <60 seconds.

ComplianceAI Enterprise — *Python · LangGraph · ChromaDB · RAG · FastAPI · Guardrails*

- Full-stack GDPR/SOX/HIPAA compliance platform with RAG document search, risk analytics, PDF report generation, and prompt injection guardrails. 40% hallucination reduction. Live on Hugging Face.

Real-Time Voice AI Agent — *Python · FastAPI · WebSockets · SSE · ElevenLabs · LLaMA 3*

- Bidirectional voice agent with STT/TTS streaming; sub-500ms time-to-first-byte (TTFB). Production-grade conversational latency over WebSockets and SSE.

Clara AI — Multimodal Clinical Receptionist — *Next.js · Groq · Vision AI · TTS · WebSockets · FastAPI*

- NHS GP triage prototype with live wound image analysis (Vision AI), real-time TTS, and auto-updating Clinical Command Center. Deployed on Vercel.

LLM-as-a-Judge Evaluation Dashboard — Python · LangChain · RAGAS · DeepEval · Langfuse · SSE

- Real-time evaluation pipeline scoring faithfulness, relevance, and context recall with automatic 70B → 8B model failover. Cut post-deployment defects 35%. Live on Hugging Face.

TECHNICAL SKILLS

Core: Python · SQL · TypeScript
LLMs & Agents: LangChain · LangGraph · OpenAI API · Anthropic API · Groq · ReAct · Tool Calling · Structured Outputs · MCP
RAG & Retrieval: RAG · Hybrid Retrieval · FAISS · ChromaDB · pgvector · BM25 · Embeddings · Reranking
LLM Evaluation: LLM-as-a-Judge · RAGAS · DeepEval · Langfuse · Prompt Engineering · Model Failover
ML & Data: Scikit-learn · Pandas · NumPy · NLP · Computer Vision · Hugging Face Transformers
Infrastructure: FastAPI · Docker · Kubernetes · AWS (EC2 · S3 · Lambda) · Oracle Cloud · GitHub Actions · CI/CD

EDUCATION

B.Tech — Electronics and Communication Engineering Expected May 2026

Ajay Kumar Garg Engineering College (AKGEC), Ghaziabad

- Technical Lead, Software Incubator — shipped 6 production AI systems used by real users across finance, healthcare, and DevOps domains.

COMMUNITY & OPEN SOURCE

Hugging Face — Active Space Developer (huggingface.co/lamEntity) 2024 – Present

- 6 publicly deployed AI Spaces: SentinelAlpha · ComplianceAI Enterprise · NeuralHeist · PersonaDoc RAG · SmartLedger · Enterprise Production Agent — all live and publicly accessible.

Google Developer Groups (GDG) — Member 2024 – Present

NVIDIA Developer Program — Member 2024 – Present

CERTIFICATIONS

Oracle Cloud Infrastructure — Generative AI Professional Oracle, 2024
Oracle Cloud Infrastructure — Data Science Professional Oracle, 2024